

Day 2 (Set B) — Fluid Mechanics I

Comprehensive Problem Set — More Practice

June 18, 2026 • 20 Questions

Focus: A second set on fluid properties, fluid statics & pressure, continuity ($Q = VA$), the Bernoulli/energy equation, Reynolds number, and laminar vs. turbulent flow — new numbers and a few new twists.

Instructions: Use only the *NCEES FE Reference Handbook*. Take $g = 9.81 \text{ m/s}^2$, $\rho_{\text{water}} = 1000 \text{ kg/m}^3$, and $\gamma_{\text{water}} = 9.81 \text{ kN/m}^3$ unless stated. Choose the answer *nearest* to your result. Two items are numeric-entry. Write units at every step.

Questions

1. A fluid has dynamic viscosity $\mu = 1.5 \times 10^{-3} \text{ Pa} \cdot \text{s}$ and density 850 kg/m^3 . Its kinematic viscosity is most nearly:
(A) $1.76 \times 10^{-6} \text{ m}^2/\text{s}$ (B) $1.28 \times 10^{-3} \text{ m}^2/\text{s}$
(C) $5.67 \times 10^5 \text{ m}^2/\text{s}$ (D) $1.76 \times 10^{-3} \text{ m}^2/\text{s}$
2. Seawater has a specific gravity of 1.03. Its specific weight is most nearly:
(A) 9.81 kN/m^3 (B) 10.1 kN/m^3
(C) 1.03 kN/m^3 (D) 10.1 N/m^3
3. A simple mercury manometer ($\text{SG} = 13.6$) open to the atmosphere shows a column height of 150 mm. The applied gauge pressure is most nearly:
(A) 2.04 kPa (B) 20.0 kPa
(C) 14.7 kPa (D) 1.47 kPa
4. A tank holds a 2.0 m layer of oil ($\text{SG} = 0.80$) floating on a 3.0 m layer of water. The gauge pressure at the bottom is most nearly:
(A) 45.1 kPa (B) 39.2 kPa
(C) 49.1 kPa (D) 29.4 kPa
5. The gauge pressure at a depth of 20 m in seawater ($\text{SG} = 1.025$) is most nearly:
(A) 196 kPa (B) 201 kPa
(C) 210 kPa (D) 20.1 kPa
6. A vertical square gate $1.0 \text{ m} \times 1.0 \text{ m}$ has its top edge 2.0 m below the water surface. The resultant hydrostatic force on the gate is most nearly:
(A) 19.6 kN (B) 24.5 kN
(C) 9.81 kN (D) 29.4 kN

7. For the gate of Problem 6, the depth of the center of pressure below the surface is most nearly:
- (A) 2.50 m (B) 2.53 m
(C) 2.83 m (D) 3.03 m
8. A block of wood of specific gravity 0.60 floats freely in water. The fraction of its volume that is submerged is most nearly:
- (A) 40% (B) 60%
(C) 80% (D) 100%
9. A hydraulic lift has piston diameters of 50 mm (input) and 300 mm (output). An input force of 200 N can support an output load of most nearly:
- (A) 1200 N (B) 7200 N
(C) 2400 N (D) 33 N
10. **(Numeric entry)** Water flows at $Q = 0.030 \text{ m}^3/\text{s}$ through a pipe of inside diameter 150 mm. Determine the mean velocity, in m/s.
11. A 300 mm pipe carrying water at 2.0 m/s splits into two identical 150 mm pipes that share the flow equally. The velocity in each branch is most nearly:
- (A) 2.0 m/s (B) 4.0 m/s
(C) 1.0 m/s (D) 8.0 m/s
12. Air ($\rho = 1.2 \text{ kg/m}^3$) flows at 10 m/s through a $0.5 \text{ m} \times 0.5 \text{ m}$ duct. The mass flow rate is most nearly:
- (A) 3.0 kg/s (B) 2.5 kg/s
(C) 6.0 kg/s (D) 1.5 kg/s
13. Water discharges from an orifice 9.0 m below the free surface of a large open tank. Neglecting losses, the jet velocity is most nearly:
- (A) 13.3 m/s (B) 9.39 m/s
(C) 18.8 m/s (D) 6.64 m/s
14. Water flows in a horizontal pipe. At point 1, $V_1 = 1.0 \text{ m/s}$ and $p_1 = 150 \text{ kPa}$; the pipe narrows so that $V_2 = 4.0 \text{ m/s}$. Neglecting losses, p_2 is most nearly:
- (A) 142.5 kPa (B) 157.5 kPa
(C) 135.0 kPa (D) 150.0 kPa
15. Water leaves a nozzle at 15 m/s. The corresponding velocity head is most nearly:
- (A) 11.5 m (B) 1.53 m
(C) 7.65 m (D) 22.9 m

16. Water flows up a constant-diameter vertical pipe. At the bottom $p_1 = 250$ kPa; the top is 10 m higher and the velocity is unchanged. Neglecting losses, the pressure at the top is most nearly:
- (A) 151.9 kPa (B) 250 kPa
(C) 348.1 kPa (D) 200 kPa
17. A pitot-static tube in a water flow reads a stagnation-minus-static pressure difference of 5.0 kPa. The flow velocity is most nearly:
- (A) 3.16 m/s (B) 10.0 m/s
(C) 2.24 m/s (D) 1.58 m/s
18. Oil ($\rho = 900$ kg/m³, $\mu = 0.018$ Pa · s) flows at 1.2 m/s in an 80 mm pipe. The Reynolds number is most nearly:
- (A) 4,800 (B) 48,000
(C) 480 (D) 960
19. Water ($\nu = 1.0 \times 10^{-6}$ m²/s) flows at 0.05 m/s in a 20 mm pipe. The Reynolds number and flow regime are:
- (A) 1,000, laminar (B) 1,000, turbulent
(C) 2,000, laminar (D) 20,000, turbulent
20. **(Numeric entry)** A rectangular duct 0.40 m $\times$ 0.20 m flows completely full. Determine the hydraulic diameter $D_H = \frac{4A}{P}$, in m.

Answer Key

Q	Ans	Q	Ans	Q	Ans
1	A	8	B	15	A
2	B	9	B	16	A
3	B	10	1.70 m/s	17	A
4	A	11	B	18	A
5	B	12	A	19	A
6	B	13	A	20	0.267 m
7	B	14	A		

Worked Solutions

$$1. \nu = \frac{\mu}{\rho} = \frac{1.5 \times 10^{-3}}{850} = \mathbf{1.76 \times 10^{-6} \text{ m}^2/\text{s}}.$$

Handbook: Fluid Properties — kinematic viscosity $\nu = \mu/\rho$

$$2. \gamma = \text{SG } \gamma_w = (1.03)(9.81) = \mathbf{10.1 \text{ kN/m}^3}.$$

Handbook: Fluid Properties — specific weight, $\gamma = \text{SG } \gamma_w$

$$3. p = \text{SG } \gamma_w h = (13.6)(9810)(0.150) = 20,012 \text{ Pa} \approx \mathbf{20.0 \text{ kPa}}.$$

Handbook: Fluid Mechanics — manometry, $p = \rho gh$

$$4. p = \gamma_{oil} h_{oil} + \gamma_w h_w = (0.80)(9.81)(2.0) + (9.81)(3.0) = 15.70 + 29.43 = \mathbf{45.1 \text{ kPa}}.$$

Handbook: Fluid Statics — layered fluids, $p = \sum \gamma_i h_i$

$$5. p = \text{SG } \gamma_w h = (1.025)(9.81)(20) = \mathbf{201 \text{ kPa}}.$$

Handbook: Fluid Mechanics — $p = \gamma h$

$$6. F_R = \gamma h_c A, \text{ with } h_c = 2.0 + 0.5 = 2.5 \text{ m and } A = (1)(1) = 1 \text{ m}^2:$$

$$F_R = (9.81)(2.5)(1) = \mathbf{24.5 \text{ kN}}.$$

Handbook: Forces on Submerged Surfaces / Center of Pressure

$$7. y_{cp} = y_c + \frac{I_{xc}}{y_c A}, \text{ with } y_c = 2.5 \text{ m, } I_{xc} = \frac{bh^3}{12} = \frac{(1)(1)^3}{12} = 0.0833 \text{ m}^4:$$

$$y_{cp} = 2.5 + \frac{0.0833}{(2.5)(1)} = 2.5 + 0.033 = \mathbf{2.53 \text{ m}}.$$

Handbook: Forces on Submerged Surfaces / Center of Pressure

$$8. \text{ A freely floating body displaces its own weight, so the submerged volume fraction equals the specific gravity: } \frac{V_{sub}}{V} = \text{SG} = \mathbf{0.60 (60\%)}. \quad \text{Handbook: Fluid Mechanics — buoyancy / flotation}$$

9. Pascal's law, equal pressure on both pistons: $F_2 = F_1 \frac{A_2}{A_1} = F_1 \left(\frac{D_2}{D_1} \right)^2 = (200) \left(\frac{300}{50} \right)^2 = (200)(36) = \mathbf{7200 \text{ N}}$.

Handbook: Fluid Statics — Pascal's law / hydraulic press

10. $A = \frac{\pi}{4} D^2 = \frac{\pi}{4} (0.150)^2 = 0.01767 \text{ m}^2$;

$$V = \frac{Q}{A} = \frac{0.030}{0.01767} = \mathbf{1.70 \text{ m/s}}.$$

Handbook: Continuity Equation, $Q = AV$

11. Total flow $Q = \frac{\pi}{4} (0.30)^2 (2.0) = 0.1414 \text{ m}^3/\text{s}$; each branch carries $0.0707 \text{ m}^3/\text{s}$ through $A = \frac{\pi}{4} (0.15)^2 = 0.01767 \text{ m}^2$:

$$V = \frac{0.0707}{0.01767} = \mathbf{4.0 \text{ m/s}}.$$

Handbook: Continuity — branching flow, $\sum Q_{in} = \sum Q_{out}$

12. $\dot{m} = \rho AV = (1.2)(0.5 \times 0.5)(10) = (1.2)(0.25)(10) = \mathbf{3.0 \text{ kg/s}}$.

Handbook: Continuity — mass flow rate, $\dot{m} = \rho AV$

13. Torricelli: $v = \sqrt{2gh} = \sqrt{2(9.81)(9.0)} = \sqrt{176.6} = \mathbf{13.3 \text{ m/s}}$.

Handbook: Bernoulli/Energy Equation — orifice discharge

14. Horizontal Bernoulli: $p_2 = p_1 + \frac{1}{2} \rho (V_1^2 - V_2^2) = 150,000 + \frac{1}{2} (1000)(1^2 - 4^2)$
 $= 150,000 - 7,500 = 142,500 \text{ Pa} = \mathbf{142.5 \text{ kPa}}$.

Handbook: Bernoulli Equation (energy along a streamline)

15. Velocity head $= \frac{V^2}{2g} = \frac{(15)^2}{2(9.81)} = \frac{225}{19.62} = \mathbf{11.5 \text{ m}}$.

Handbook: Bernoulli/Energy Equation — velocity head

16. Constant diameter $\Rightarrow$ velocity head cancels: $p_2 = p_1 - \gamma(z_2 - z_1) = 250,000 - (9810)(10)$
 $= 250,000 - 98,100 = 151,900 \text{ Pa} \approx \mathbf{151.9 \text{ kPa}}$.

Handbook: Bernoulli/Energy Equation — elevation head

17. Pitot-static: $V = \sqrt{\frac{2\Delta p}{\rho}} = \sqrt{\frac{2(5000)}{1000}} = \sqrt{10} = \mathbf{3.16 \text{ m/s}}$.

Handbook: Fluid Mechanics — pitot tube / stagnation pressure

18. $Re = \frac{\rho V D}{\mu} = \frac{(900)(1.2)(0.080)}{0.018} = \mathbf{4800}$. (Just turbulent, $Re > 4000$.)

Handbook: Reynolds Number, $Re = \rho V D / \mu$

19. $Re = \frac{VD}{\nu} = \frac{(0.05)(0.020)}{1.0 \times 10^{-6}} = \mathbf{1000}$; since $Re < 2100$ the flow is **laminar**.

Handbook: Reynolds Number — flow regimes

20. $D_H = \frac{4A}{P} = \frac{4(0.40 \times 0.20)}{2(0.40 + 0.20)} = \frac{0.32}{1.2} = \mathbf{0.267 \text{ m}}$.

Handbook: Reynolds Number — hydraulic diameter $D_H = 4A/P$

All numeric results independently verified. Keep practicing locating each equation in the NCEES FE Reference Handbook.